THE LOCUS OF POINTS OF A FIBRE WHERE A FINITE TYPE MODULE IS FLAT IS OPEN IN THE FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The local structure statement for a finite type flat sheaf, and permanence of flatness. Same chapter flat.tex, label proposition-finite-type-flat-at-point (proof line 2690); Morphisms, label morphisms-lemma-flat-permanence (proof line 2695). The first provides an elementary etale neighbourhood and an open subscheme on which the pullback is flat, and the second shows that flatness of the pullback at a point of the fibre implies flatness of the original module there.

Source reference. Original lemma 02JZ: sources/stacks-chow/tags.txt, source lines 3354–3354. Navigation pointer for \ref{morphisms-lemma-flat-permanence} used inside the counted span (line 2695); the argument that consumes it is retained in the span.

Source reference. Original proposition 05I5: sources/stacks-flat/pinned-original-source.tex, source lines 2477–2477. Navigation pointer for \ref{proposition-finite-type-flat-at-point} used inside the counted span (line 2690); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module of finite type. Let $s \in S$. Then the set*

$$\{x \in X_s \mid \mathcal{F} \text{ flat over } S \text{ at } x\}$$

is open in the fibre X_s.

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Proof. Suppose $x \in U$. Choose an elementary étale neighbourhood $(S', s') \to (S, s)$ and open $V \subset X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'})$ as in Proposition 05I5. Note that $X_{s'} = X_s$ as $\kappa(s) = \kappa(s')$. If $x' \in V \cap X_{s'}$, then the pullback of $\mathcal{F}$ to $X \times_S S'$ is flat over S' at x'. Hence $\mathcal{F}$ is flat at x' over S, see Morphisms, Lemma 02JZ. In other words $X_s \cap V \subset U$ is an open neighbourhood of x in U. $\square$